Daniel Quillen's Notebooks: Statements Proved in Lean

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Abstract

Daniel Quillen's research notebooks, from 1968 to 2003, are online at the Clay Mathematics Institute. This document picks out ten precise statements in them, in general topology, linear algebra, group theory, representation theory, category theory, combinatorics and analysis on the torus, and proves them. Each proof follows a proof checked in the Lean 4 proof assistant [2], with its mathlib library [3], and is written to be read without Lean. For each statement, the document says where it comes from, what it means, how it is proved, what is not formalized, and what is known of its presence in the literature. All ten statements are correct; three hold under weaker hypotheses than those of the page or of the expected proof, one fails for $p = 2$, as the page suggests, and six are classical but absent from mathlib.

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Introduction

The notebooks. Daniel Quillen (1940–2011) kept working notebooks all his life, dated, in which he tries out ideas, redoes computations and notes statements. The Clay Mathematics Institute has digitized them and publishes them at `https://www.claymath.org/library/Quillen/`: several hundred notebooks, from 1968 to 2003, one PDF per notebook. This document cites each notebook by its year and number, in the form "notebook 1968-14, p. 27", where the page is that of the PDF; each citation points to the Clay PDF.

The rule on rights. The notebooks are protected by copyright. This document quotes almost nothing from them, a few words at most; it states each result in its own terms and refers to the page of the facsimile, which anyone can consult. The transcriptions made for this work remain private.

The method. The notebooks were read in three stages. First, the precise statements are located: lemmas, propositions, identities, questions asked. Each statement is then the object of a literature search. Most of what one reads in the notebooks is found again in Quillen's published papers, on homotopical algebra, group cohomology and algebraic K-theory, cobordism, superconnections, determinants and cyclic cohomology; those statements are not taken up here. Finally, only the precise statements that the search did not find in print, or whose printed form does not match the page, are formalized. Four statements of this document are in that case. The others, in notebooks 1968-1, 1968-13 (restriction of representations), 1968-15 and 16, 1970-15, 1971-4 and 1983-4, are classical and found in the literature; they are formalized because the pages rely on them and mathlib does not have them. The red lozenge ♦, in the title of a section, marks the four notebooks in which a proved statement is a candidate lead: the search did not find it in print in this form, which does not establish it as new.

What Lean guarantees. Each section cites, in a footnote to its title, the Lean file that contains its proofs, in the `lean/Quillen/` library of the project, one file per notebook. Lean mechanically checks that a statement holds under the hypotheses as written. It does not guarantee that the statement faithfully renders the page, nor that it is new. A statement that the page formulates carries the name of Quillen, the notebook and the page; when the proof establishes more, we say "after"; a form that the edition adds (a generalization, an answer to a question, an intermediate lemma) is called the edition's statement.

Organization. One section per statement, in the chronological order of the notebooks. Each presents the page in its context, gives the statement with its intuition and its proof, says what Lean checks and what is not proved, then the state of the literature search.

Summary table

Notebook	Statement	Verdict	Lean file	§
1968-1, p. 4	A simple module is determined by its trace	classical, found; in every characteristic, over an algebraically closed field	`Notebook1968_1`	1

Notebook	Statement	Verdict	Lean file	§
1968-13, pp. 64–69	The space $X \cup_\Phi \{\infty\}$ and the smash lemma	space found (Beer–Vipera); smash lemma more general (neither local compactness nor separation)	`Notebook1968_13`	2
1968-13, p. 4	Restriction of representations is full if and only if f is surjective	classical, found; equivalence if and only if f is bijective	`Notebook1968_13Rep`	3
1968-14, pp. 27–28	Compact preimages of closed sets away from A	confirmed; more general (neither local compactness, nor σ-compactness, nor A closed)	`Notebook1968_14`	4
1968-15, p. 28; 1968-16, pp. 1–2	$E(n, A)$ perfect for $n \geq 3$; Whitehead's lemma	classical, found; over any unital ring	`Notebook1968_15`, `_16`	5
1970-15, pp. 21–22	Möbius function of the lattice of subspaces	classical, found; Gauss's binomial theorem	`Notebook1970_15`	6
1971-4, pp. 24–25	A commutative monoid in its Grothendieck group: flat functor	classical, found; flat and final	`Notebook1971_4`	7
1971-11, pp. 68–71	Automorphisms of order p trivial on Ω_1; quotient $P/\Omega_1(P)$ of a p-central p-group	confirmed; more general (neither finiteness nor bijectivity); false for $p = 2$; quotient: found (González-Sánchez–Weigel)	`Notebook1971_11`, `Central`	8
1983-4, pp. 71–72	Cohomological equation of $\theta\partial_x - \partial_y$ on the torus	classical, found; both directions, for every irrational θ	`Notebook1983_4`	9
1983-10, p. 31	Traces of odd powers of $P - Q$	question of the page; answer in finite dimension, over any commutative ring	`Notebook1983_10`	10

1 Notebook 1968-1: a simple module is determined by its trace[1]

Context. notebook 1968-1, dated January 1968, is mostly about representations of semisimple Lie algebras. On PDF page 4, Quillen isolates a basic statement: an irreducible representation of a finite-dimensional algebra over $\mathbb{C}$ is determined by its trace function $a \mapsto \operatorname{tr}(a \mid S)$. His sketch: the trace vanishes on the radical, whose elements act nilpotently; by Wedderburn, the semisimple quotient is a product of simple algebras, and the trace of S is nonzero on the factor of S and zero on the others. A remark that the trace equals $\dim S$ at 1 is crossed out, rightly so in characteristic p, where $\dim S$ can vanish in the field.

[1] Proof checked: `lean/Quillen/Notebook1968_1.lean`.

Intuition. If $S \not\simeq T$, it suffices to find an element of A acting on S as a projector of rank one and on T as zero: its traces are 1 and 0, in every characteristic. Jacobson's density theorem provides such an element, because the endomorphisms of $S \times T$ are diagonal and scalar.

Theorem 1.1 (Quillen, notebook 1968-1, p. 4)**.** *Let k be an algebraically closed field of arbitrary characteristic, A a k-algebra, and S, T two simple A-modules of finite dimension over k. If $\operatorname{tr}(a \mid S) = \operatorname{tr}(a \mid T)$ for all $a \in A$, then $S \simeq T$. More precisely, if $S \not\simeq T$, there exists $a \in A$ with $\operatorname{tr}(a \mid S) = 1$ and $\operatorname{tr}(a \mid T) = 0$. When A is finite-dimensional, as in the notebook, the finite-dimensionality hypothesis on S and T is automatic.*

Lemma 1.2 (Schur)**.** *An A-linear map between non-isomorphic simple modules is zero. If k is algebraically closed and S is of finite dimension over k, every A-linear endomorphism of S is a scalar multiplication.*

Proof. The kernel and the image are submodules, so a nonzero map between simple modules is bijective. An endomorphism g of S has an eigenvalue $c \in k$; $g - c$ is A-linear and not injective, hence zero. □

Proof of the theorem. The module $S \times T$ is semisimple, being the sum of the images of the two injections. By Lemma 1.2, if $S \not\simeq T$, an A-linear endomorphism g of $S \times T$ has the form $g(s,t) = (c\,s, d\,t)$ with $c, d \in k$. Let $v \neq 0$ in S, φ a linear form with $\varphi(v) = 1$, and $\psi = \varphi(\cdot)\,v$, of trace 1. The map $f(s,t) = (\psi(s), 0)$ commutes with all these g; since $S \times T$ is finitely generated over its endomorphism ring (it already is over k), the density theorem gives $a \in A$ acting as f. Then $\operatorname{tr}(a \mid S) = \operatorname{tr}\psi = 1$ and $\operatorname{tr}(a \mid T) = 0$, which contradicts the equality of traces. If A is finite-dimensional, a simple module is cyclic, hence finite-dimensional over k. □

Remark. The proof avoids the radical and Wedderburn–Artin, which Quillen uses; the two routes give the same element, a matrix unit e_{11} of the block of S. The finiteness of A is not used, only that of S and T.

Counterexample 1.3. *The hypothesis "algebraically closed" cannot be dropped in characteristic p. Let L/k be a purely inseparable extension of degree p and $A = L \times L$; the two simple modules $L \times 0$ and $0 \times L$ are not isomorphic, and their traces, $(x,y) \mapsto \operatorname{Tr}_{L/k}(x)$ and $(x,y) \mapsto \operatorname{Tr}_{L/k}(y)$, are both zero. Likewise, over an algebraically closed field of characteristic p, the trace does not determine semisimple modules: $S^{\oplus p}$ and the zero module have the same trace.*

Verification in Lean. The theorem is `nonempty_linearEquiv_of_trace_eq`, and `nonempty_linearEquiv_of_trace_eq'` for A finite-dimensional; the element with traces 1 and 0 is `exists_trace_eq_one_trace_eq_zero`. Lemma 1.2 is `eq_zero_of_isEmpty_linearEquiv` and `exists_eq_smul_of_isAlgClosed`, the semisimplicity of $S \times T$ is `isSemisimpleModule_prod`, in `Notebook1968_1.lean`. Density is mathlib's `Module.Finite.toModuleEnd_moduleEnd_surjective`.

Not formalized: the counterexample above, and the linear independence of the traces of pairwise non-isomorphic simple modules, which follows from the same argument.

Literature. found in the literature: Curtis–Reiner, *Representation Theory of Finite Groups and Associative Algebras* (1962), the chapter on characters of algebras; Lam, *A First Course in Noncommutative Rings*, the chapter on representations. mathlib has the density theorem and Wedderburn–Artin over an algebraically closed field, but not this statement; it could be proposed for mathlib, and the edition does not present it as new.

2 Notebook 1968-13: the space $X \cup_\Phi \{\infty\}$ of a family of supports, and the smash lemma ♦[2]

Context. In December 1968, notebook 1968-13 relates supports and bordism. On page 64, Quillen takes on a space X a family of supports Φ: a family of closed sets, stable under finite unions and under passage to closed subsets. On page 65, he forms the space $X \cup_\Phi \{\infty\}$, in which X is open and the neighbourhoods of ∞ are the sets $(X \setminus Z) \cup \{\infty\}$, $Z \in \Phi$. For Φ the family of closed sets, ∞ is an isolated point; for Φ the family of compact sets, one recovers the Alexandroff compactification X^+. A crossed-out passage on page 64 notes that the space is Hausdorff if Φ contains a neighbourhood of each point. On pages 67–69, after his own correction ("Correction": he had defined the smash product wrongly, it being the quotient of $X \times Y$ by the wedge), he states and proves that $X^+ \wedge (Y \cup_\Phi \{\infty\}) = (X \times Y) \cup_\Psi \{\infty\}$, where Ψ is the family of closed subsets of $X \times Y$ contained in some $X \times F$, $F \in \Phi$, and proper over F.

Intuition. This is the argument of the page. A neighbourhood of ∞ in the smash product is the image of an open set containing the wedge $X^+ \times \{\infty\} \cup \{\infty\} \times (Y \cup_\Phi \{\infty\})$; its complement is a closed set C of $X \times Y$, closed in the whole product. Since X^+ is compact, the tube lemma keeps C away from $X^+ \times \{\infty\}$, that is, places it inside some $X \times F$; and C is closed in $X^+ \times F$ (figure 1).

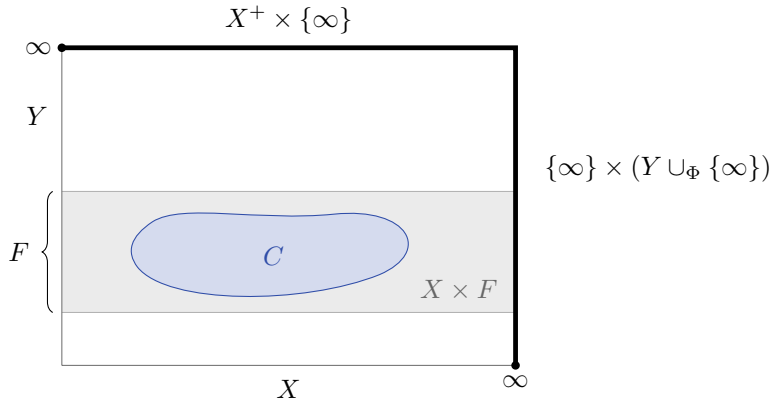


Figure 1: The product $X^+ \times (Y \cup_\Phi \{\infty\})$, with X^+ on the horizontal axis and $Y \cup_\Phi \{\infty\}$ on the vertical axis. The wedge, in thick line, is the union of the upper edge $X^+ \times \{\infty\}$ and the right edge $\{\infty\} \times (Y \cup_\Phi \{\infty\})$. The complement of a neighbourhood of ∞ in the smash product is a closed set C of $X \times Y$ (in blue). The tube lemma, applied to the compact set $X^+ \times \{\infty\}$, places it in a strip $X \times F$ with $F \in \Phi$ (in grey), and C is closed in $X^+ \times F$: it does not accumulate on the right edge.

Theorem 2.1 (Quillen, notebook 1968-13, pp. 64–69)**.** *Let X, Y be topological spaces and Φ a family of supports on Y.*

a) X is an open subset of $X \cup_\Phi \{\infty\}$ and $\{\infty\}$ is closed. If X is Hausdorff, $X \cup_\Phi \{\infty\}$ is Hausdorff if and only if every point of X has a neighbourhood in Φ.

b) For Φ the family of compact closed sets, $X \cup_\Phi \{\infty\}$ is homeomorphic to X^+; for Φ the family of closed sets, ∞ is isolated.

[2] Proof checked: `lean/Quillen/Notebook1968_13.lean`.

c) Let Ψ be the family of closed subsets A of $X \times Y$ contained in some $X \times F$, $F \in \Phi$, and closed in $X^+ \times Y$. The map $X^+ \times (Y \cup_\Phi \{\infty\}) \to (X \times Y) \cup_\Psi \{\infty\}$, which is the identity on $X \times Y$ and sends the wedge to ∞, is a quotient map: $(X \times Y) \cup_\Psi \{\infty\}$ is the smash product $X^+ \wedge (Y \cup_\Phi \{\infty\})$ with the quotient topology.

Quillen assumes X locally compact, which makes X^+ Hausdorff; the proof uses only the compactness of X^+, and c) holds without any hypothesis. The family Ψ is taken in the form that the proof on the page uses (closed in $X^+ \times F$); the equivalence with "proper over F", which requires X^+ Hausdorff, is not formalized.

Proof. a) Separating $x \in X$ from ∞ requires an open set $U \ni x$ disjoint from some $X \setminus Z$, $Z \in \Phi$, that is, $U \subset Z$: Z is a neighbourhood of x. Conversely, the interior of Z and $(X \setminus Z) \cup \{\infty\}$ separate x and ∞.

b) An open set containing ∞ has, in both topologies, a complement in X that is closed and compact, or contained in a compact closed set, which amounts to the same thing. For the closed sets, X itself is in Φ, and $\{\infty\}$ is open.

c) The map is surjective. For a set U not containing ∞, its preimage is the image of $U \cap (X \times Y)$ under the open embedding $X \times Y \to X^+ \times (Y \cup_\Phi \{\infty\})$. For $U \ni \infty$, the preimage is the complement of the image of the closed set $C = (X \times Y) \setminus U$, and everything comes down to seeing that this image is closed if and only if $C \in \Psi$. If it is closed, the tube lemma, applied to the compact set $X^+ \times \{\infty\}$, gives $F \in \Phi$ with $C \subset X \times F$, and the preimage under $\mathrm{id} \times \iota : X^+ \times Y \to X^+ \times (Y \cup_\Phi \{\infty\})$ shows that C is closed in $X^+ \times Y$. Conversely, the image of C is contained in the closed set $X^+ \times F$, which does not meet $\{\infty\}$, and it is there the image of a closed set under an embedding. □

Verification in Lean. `SupportFamily` and `Extension` define the family and the space; a) is `isOpenEmbedding_ι`, `isClosed_infty` and `t2Space_iff`; b) is `homeomorphOnePoint`, to mathlib's compactification `OnePoint X`, `isOpen_infty_closedSupports` and `homeomorphSum`, to the sum of X and a point; c) is `isQuotientMap_smashMap`, with the family `smashSupports` and the key lemma `isClosed_image_e_iff`, in `Notebook1968_13.lean`.

Not formalized: $\Gamma(X^+, j_! F) = \Gamma_\Phi(X, F)$ (p. 65), the exact sequence that follows from it (p. 66), and the application to bordism (pp. 66–69).

Literature. the space is found in the literature, the smash lemma is not. G. Beer and M. C. Vipera associate to every family of ideals of subsets with a closed base of a space an extension of that space, of which the Alexandroff compactification is the prototype [5]; a family of supports generates such an ideal, and the one-point extension corresponding to it is $X \cup_\Phi \{\infty\}$. G. Beer was already studying the one-point extension of a space equipped with a bornology [4], a setting that goes back to S.-T. Hu [6]. These references rest on the zbMATH Open reviews (Zbl 1259.54008 and 1171.54012); the full text has not been read, and it has not been checked that the separation criterion a) appears there in this form. Supports and Γ_Φ are due to Godement (1958) and to Bredon (*Sheaf Theory*), whose book has not been combed page by page. The smash lemma has not been found in print.

3 Notebook 1968-13: when restriction of representations is full[3]

Context. notebook 1968-13 opens, in autumn 1968, with a Whitehead theorem for small categories: a functor $f : \mathcal{C}_1 \to \mathcal{C}_2$ should be a weak equivalence as soon as pullback along f induces an equivalence between derived categories of locally constant coefficient systems. In the proof of the implication "(i) $\Rightarrow$ (ii)" (PDF, p. 4), Quillen reduces to the case of a group homomorphism and needs a small lemma, which he states in parentheses: restriction of modules along a group homomorphism is fully faithful exactly when the homomorphism is surjective, and an equivalence exactly when it is bijective. He justifies surjectivity by comparing the invariants of the permutation module $\mathbb{Z}[G/f(H)]$ under the two groups, and injectivity by noting that the kernel acts trivially on every restricted module. On the page, the letters G and H are swapped between the statement and the rank argument.

Intuition. A map between restrictions commutes with the action of the image $f(H)$, not necessarily with that of G; it suffices to find a module where the gap shows. In $k[G/f(H)]$, the basis class is fixed by $f(H)$ and by it alone: the projection onto the line it spans commutes with H, and it commutes with G only if G fixes this class, that is, if $f(H) = G$. For injectivity, the regular representation $k[H]$ is faithful, whereas the kernel of f acts trivially on everything that comes from G.

Theorem 3.1 (Quillen, notebook 1968-13, p. 4)**.** *Let $f : H \to G$ be a group homomorphism and k a nonzero commutative ring. The restriction functor $\mathrm{Rep}_k(G) \to \mathrm{Rep}_k(H)$ is full if and only if f is surjective; it is an equivalence if and only if f is bijective.*

Proof. Restriction is always faithful: it does not change the underlying linear map. If f is surjective, an H-linear map between restrictions commutes with every $g = f(h)$, hence is G-linear.

Suppose restriction is full. Let $V = k[G/f(H)]$, c_0 the class of $f(H)$, and $p : V \to V$ the map $x \mapsto x(c_0)\, c_0$. For $h \in H$, $f(h)$ and $f(h)^{-1}$ fix c_0, so $p(f(h)x) = f(h)\, p(x)$: p is H-linear. It therefore comes from a G-linear map, which is still p. For $g \in G$, we have $p(gc_0) = g\, p(c_0) = gc_0 \neq 0$ since k is nonzero, so $gc_0 = c_0$, that is, $g \in f(H)$.

If f is bijective, with inverse f^{-1}, every representation ρ of H is the restriction of $\rho \circ f^{-1}$; restriction is then full, faithful and essentially surjective. Conversely, suppose it is essentially surjective and let $h \in \ker f$. The regular representation $k[H]$ is isomorphic to the restriction of a representation W of G; h acts on W by $f(h) = 1$, hence trivially, and by transport it fixes $\delta_1 \in k[H]$. But $h\, \delta_1 = \delta_h$, whence $h = 1$. □

Verification in Lean. In `Notebook1968_13Rep.lean`, restriction is mathlib's functor `Rep.resFunctor f`. The fullness criterion is `full_resFunctor_iff`, whose forward direction is `surjective_of_full`, with the projection `proj` onto the basis class of the module `cosetRep`; the equivalence criterion is `isEquivalence_resFunctor_iff`, with `injective_of_essSurj` (tested on `regRep`) and `essSurj_of_bijective`. The representations live in the universe $\max(u, v_1, v_2)$, large enough to contain the test objects.

Quillen's case is $k = \mathbb{Z}$; the rank argument on invariants is replaced by the projection, which is equivalent to it (a map from the trivial representation is an invariant). Not formalized: the Whitehead theorem itself, which requires the derived categories of locally constant coefficients.

[3] Proof checked: `lean/Quillen/Notebook1968_13Rep.lean`.

Literature. folklore, found in the literature: inflation along a quotient $G \to G/N$ identifies G/N-modules with the G-modules on which N acts trivially, a full subcategory. mathlib has the direction "surjective ⇒ full" (`Rep.full_res`, and `Action.full_res` for actions) but neither the converse nor the equivalence criterion. They are proposed for mathlib, and the edition does not present them as new.

4 Notebook 1968-14: maps whose preimages of closed sets away from A are compact ♦[4]

Context. notebook 1968-14, from December 1968, looks for a geometric description of homology by bordism classes of maps of oriented manifolds. For the homology of a space X relative to a closed set A, the cycles proposed on pages 26–27 are maps $f : V \to X \setminus A$ such that $f^{-1}(B)$ is compact for every closed set B of X disjoint from A. Quillen notes that, if X is compact, this condition amounts to saying that f is proper as a map to $X \setminus A$: a closed set of X disjoint from A is then a compact set of $X \setminus A$, and conversely.

The page. On page 27, he states a proposition on inductive limits taken over the compact subsets of A. The heart of the proof is the following point: if f satisfies the condition, the intersection of the closure of $f(V)$ with A is compact. He argues by contradiction: a non-compact closed set contains a discrete sequence (a_n); one chooses an exhausting sequence (Q_n) of compact subsets of V, then points $x_n \notin Q_n$ whose images $f(x_n)$ are close enough to the a_n to form a discrete set. This set is closed, disjoint from A, and its preimage is contained in no Q_n: it is not compact. At the top of page 28, a remark draws from the proof the equivalence of the condition with two properties: f is proper, and the closure of $f(V)$ in X is compact. This is the remark proved here. Its spaces are manifolds; the proof on the page uses an exhaustion of V by compact sets, hence the σ-compactness of V.

Intuition. A proper map to $X \setminus A$ can still "escape to infinity" in two ways, seen from X: by approaching A, which the condition allows, or by leaving every compact subset of X, which it forbids. The condition says exactly that f escapes only towards A (figure 2). The compactness of the closure of $f(V)$ rules out the second escape; properness to $X \setminus A$ says that the first is the only one.

Theorem 4.1 (after Quillen, notebook 1968-14, p. 28)**.** *Let X be a metrizable space, A a subset of X, V a topological space and $f : V \to X$ a continuous map with values in $X \setminus A$. The following conditions are equivalent:*

1. *for every closed set B of X disjoint from A, the subset $f^{-1}(B)$ of V is compact;*
2. *f is proper as a map from V to $X \setminus A$, and the closure of $f(V)$ in X is compact.*

The hypotheses are weaker than those of the page: X is not assumed locally compact, A is not assumed closed, and V is arbitrary, without σ-compactness. The proof below replaces the exhaustion of V by sequences of values of f. Only the metrizability of X is used, in the direction (1) ⇒ (2), and only so that one can argue with sequences.

Proof. (2) ⇒ (1). Let K be the closure of $f(V)$, which is compact. For a closed set B disjoint from A, the set $B \cap K$ is compact, as a closed subset of a compact set, and contained in $X \setminus A$;

[4] Proof checked: `lean/Quillen/Notebook1968_14.lean`.

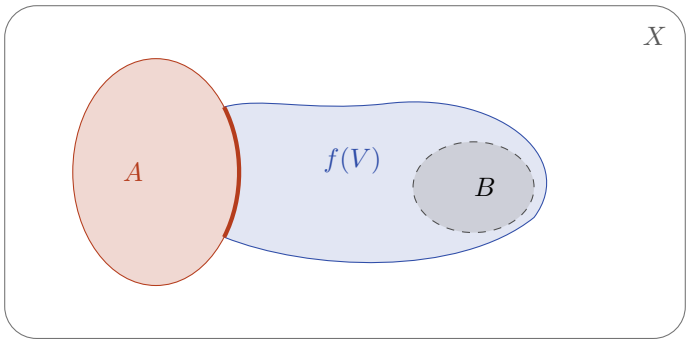


Figure 2: The conditions of Theorem 4.1. The image $f(V)$ (in blue) avoids A (in red) but approaches it: its closure meets A, here along the thick arc. Condition (1) requires the preimage of every closed set B disjoint from A (in grey) to be compact: f may escape towards A, and only towards A. Condition (2) says the same thing differently: f is proper to $X \setminus A$, and the closure of $f(V)$ is compact.

it is therefore a compact subset of $X \setminus A$. Since $f(V) \subset K$, we have $f^{-1}(B) = f^{-1}(B \cap K)$, the preimage of a compact set under a proper map: it is compact. This direction uses no hypothesis on X.

A lemma on sequences. Assume (1). Let (v_n) be a sequence in V; we show that the sequence $(f(v_n))$ has a cluster point in X. Otherwise, let B be the set of its values. It is closed: a point y of the closure of B not in B would be a cluster point of the sequence, because every neighbourhood of y, minus the finitely many values $f(v_0), \dots, f(v_{N-1})$ (which form a closed set not containing y), still meets B, hence contains some $f(v_n)$ with $n \geq N$. It is disjoint from A, since f avoids A. By (1), $f^{-1}(B)$ is compact; it contains all the v_n, which therefore have a cluster point w. By continuity, $f(w)$ is a cluster point of $(f(v_n))$: a contradiction.

(1) $\Rightarrow$ compact closure. Equip X with a metric. In a metric space, a subset is compact if and only if every sequence in it has a subsequence converging in the subset. Let (z_n) be a sequence in the closure K of $f(V)$. Choose $v_n \in V$ with $d(z_n, f(v_n)) < 1/(n+1)$. By the lemma, $(f(v_n))$ has a cluster point y, hence a subsequence $(f(v_{\varphi(k)}))$ tending to y (a metric space has countable neighbourhood bases). Since $\varphi(k) \geq k$, the distance from $z_{\varphi(k)}$ to $f(v_{\varphi(k)})$ is bounded by $1/(k+1)$, and $z_{\varphi(k)}$ also tends to y. Finally y is a limit of points of $f(V)$, hence in K.

(1) $\Rightarrow$ properness. The space $X \setminus A$ is metrizable, hence Hausdorff and compactly generated; for such a target space, a continuous map is proper if and only if the preimage of every compact set is compact. Let C be a compact subset of $X \setminus A$. Seen in X, it is compact, hence closed (X is Hausdorff), and it is disjoint from A. Its preimage is compact by (1). □

The proof on the page goes through the intersection of the closure of $f(V)$ with A; the present one shows directly that the whole closure is compact, which is what the remark on the page asserts. The choice of a discrete sequence, which Quillen makes in A and approximates by values of f, is made here directly among the values of f, which dispenses with the exhaustion of V.

Example 4.2. Let $X = \mathbf{R}$, $A = \{0\}$ and f the inclusion of $V =]0, 1[$. The closed set $B = [1/2, 1]$ does not meet A, and $f^{-1}(B) = [1/2, 1[$ is not compact: condition (1) fails, and f is not proper to $\mathbf{R} \setminus \{0\}$, although the closure $[0, 1]$ of $f(V)$ is compact. With $V =]0, 1]$, on the contrary, $f^{-1}(B) = B \cap [0, 1]$ is compact for every closed set B avoiding 0: both conditions hold, and

the map escapes towards $A = \{0\}$ and only towards it. With $A = \emptyset$, condition (1) applied to $B = X$ says that V is compact, and (2) says so too.

Verification in Lean. The file `Notebook1968_14.lean` proves Theorem 4.1 under the name `preimage_compact_iff`, for X equipped with the class `MetrizableSpace` and f restricted to the target A^c by `Set.codRestrict`. The lemma on sequences is `exists_mapClusterPt`, relying on `mapClusterPt_of_mem_closure_range`; the compactness of the closure is `isCompact_closure_range`, properness is `isProperMap_codRestrict`, by mathlib's criterion `isProperMap_iff_isCompact_preimage`; the converse is `isCompact_preimage_of_isProperMap`, stated for an arbitrary space X. The axioms used are the three usual axioms of mathlib (`propext`, `Classical.choice`, `Quot.sound`).

Not proved here: the proposition of page 27 on inductive limits of homology, whose statement is the starting point; the description of homology by bordism; the question whether the metrizability of X can be replaced by a weaker hypothesis (a sequential space, for instance).

Literature. the search did not find this statement in print in this form. It is elementary and close to classical results on proper maps (Bourbaki, *Topologie générale*, chapter I, § 10); the setting, relative homology by bordism, is connected with Quillen's work on cobordism published in 1969–1971. The edition does not present it as new. It is not a lead.

5 Notebooks 1968-15 and 1968-16: perfect subgroups, the elementary group and Whitehead's lemma[5]

Context. notebook 1968-15 and notebook 1968-16 contain the fair copy of the "plus" construction. In § 1 of the first, "Killing a perfect subgroup of the fundamental group", Proposition 1.2 kills a perfect subgroup E of $\pi_1 X$; its proof (PDF, pp. 28–29) first replaces E by the normal subgroup N it generates, still perfect because $[N, N]$ is normal and contains $[E, E] = E$. In § 2 of the second, "The K-theory associated to a ring" (p. 1), the construction is applied to the subgroup $E(A)$ of $GL(A)$ generated by the elementary matrices, which is perfect (Quillen refers to Bass); page 2 uses that $E(A)$ is normal in $GL(A)$, which rests on Whitehead's lemma. The ring A is arbitrary there, "not necessarily commutative".

Intuition. The elementary matrix $e_{ij}(a) = 1 + a\, e_{ij}$ is a commutator as soon as a third index k is available: $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$. For Whitehead's lemma, the block unipotent matrices $\left(\begin{smallmatrix} 1 & B \\ 0 & 1 \end{smallmatrix}\right)$ are products of elementary matrices, one for each coefficient of B, and three of them build the antidiagonal matrix $\left(\begin{smallmatrix} 0 & \theta \\ -\theta^{-1} & 0 \end{smallmatrix}\right)$.

Theorem 5.1 (Quillen, notebooks 1968-15, p. 28, and 1968-16, pp. 1–2)**.** *Let G be a group and A a unital ring.*

a) The normal subgroup generated by a perfect subgroup of G is perfect.

b) If i, k, j are pairwise distinct, $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$; in particular, the subgroup $E(n, A)$ of invertible matrices generated by the $e_{ij}(a)$ is perfect as soon as $n \geq 3$.

c) (Whitehead's lemma.) For θ invertible in $M_n(A)$, the matrix $\mathrm{diag}(\theta, \theta^{-1})$ is in $E(2n, A)$; for θ, η invertible, $\mathrm{diag}(\theta\eta\theta^{-1}\eta^{-1}, 1)$ is too.

[5] Proof checked: `lean/Quillen/Notebook1968_15.lean`.

Proof. a) Let H be perfect and N the normal subgroup it generates. The commutator subgroup $[N,N]$ is normal and contains $[H,H]=H$, hence contains N, so $[N,N]=N$.

b) Put $X = a\,e_{ik}$ and $Y = b\,e_{kj}$. Then $X^2 = Y^2 = YX = 0$, $XY = ab\,e_{ij}$, and XY annihilates X and Y on the right; one checks that $(1+X)(1+Y)(1-X)(1-Y) = 1+XY$. For $n \geq 3$, every generator $e_{ij}(a)$ is the commutator of $e_{ik}(a)$ and $e_{kj}(1)$, for a third index k.

c) $\left(\begin{smallmatrix}1 & B\\ 0 & 1\end{smallmatrix}\right)\left(\begin{smallmatrix}1 & C\\ 0 & 1\end{smallmatrix}\right) = \left(\begin{smallmatrix}1 & B+C\\ 0 & 1\end{smallmatrix}\right)$, and $\left(\begin{smallmatrix}1 & a\,e_{ij}\\ 0 & 1\end{smallmatrix}\right)$ is elementary; likewise for lower triangular matrices. A block computation gives

$$\begin{pmatrix}1 & \theta\\ 0 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ -\theta^{-1} & 1\end{pmatrix}\begin{pmatrix}1 & \theta\\ 0 & 1\end{pmatrix} = \begin{pmatrix}0 & \theta\\ -\theta^{-1} & 0\end{pmatrix},$$

and the product of this matrix with the same one for $\theta = -1$ is $\mathrm{diag}(\theta,\theta^{-1})$. Finally

$$\mathrm{diag}(\theta\eta\theta^{-1}\eta^{-1},1) = \mathrm{diag}(\theta,\theta^{-1})\,\mathrm{diag}(\eta,\eta^{-1})\,\mathrm{diag}((\eta\theta)^{-1},\eta\theta).$$

□

Verification in Lean. a) is `normalClosure_isPerfect`, b) is `commutator_transvectionUnit` and `elementary_isPerfect`, with the ring identity `commutator_aux`, in `Notebook1968_15.lean`; c) is `exists_mem_elementary_diag` and `exists_mem_elementary_commutator`, with `upper_mem_elementary`, `lower_mem_elementary` and `upper_lower_upper`, in `Notebook1968_16.lean`. The matrices are the units of the ring `Matrix n n A`, for an arbitrary ring A.
Not formalized: the stable statements ($E(A)$ is normal in $GL(A)$, and $E(A)$ is the commutator subgroup $[GL(A),GL(A)]$), which require the inductive limit over n, and the "plus" construction itself.

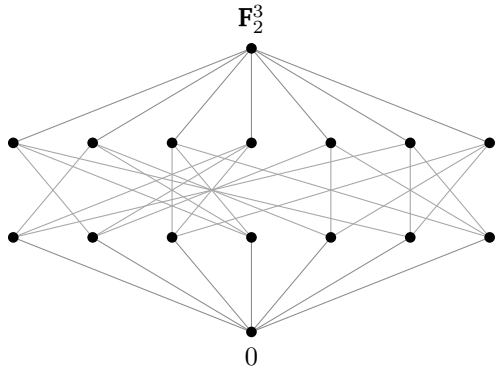
Literature. found in the literature: Milnor, *Introduction to Algebraic K-Theory* (1971), § 3 and Lemma 3.1; Rosenberg, *Algebraic K-Theory and its Applications*, § 2.1. mathlib has perfect groups (`Group.IsPerfect`) but none of these three statements. They are proposed for mathlib, and the edition does not present them as new.

6 Notebook 1970-15: the Möbius function of the lattice of subspaces[6]

Context. In September 1970, notebook 1970-15 compares the cohomology of a group in a topos with that of its points. On pages 21–22, Quillen inverts Euler characteristics over the isotropy types of an action, with Möbius coefficients, and checks the formula for an elementary abelian group acting on a point. In rank 3, the check reduces to the identity $1-(q^2+q+1)+q(q^2+q+1)-q^3=0$: the alternating sum, over the subspaces W of a space of dimension n over $\mathbf{F}_q$, of $(-1)^{\dim W} q^{\binom{\dim W}{2}}$ is zero for $n \geq 1$, since $(-1)^k q^{\binom{k}{2}}$ is the value of the Möbius function of the lattice of subspaces on an interval of rank k. Figure 3 shows the case $n=3$, $q=2$.

Intuition. The number of subspaces of dimension k is the Gaussian binomial coefficient $\left[\begin{smallmatrix}n\\ k\end{smallmatrix}\right]_q$, and these coefficients satisfy Gauss's binomial theorem $\prod_{i<n}(1+q^i t) = \sum_k q^{\binom{k}{2}} \left[\begin{smallmatrix}n\\ k\end{smallmatrix}\right]_q t^k$. For $t=-1$, the factor $i=0$ of the product vanishes.

[6] Proof checked: `lean/Quillen/Notebook1970_15.lean`.



dimension, number × value

$k = 3$: $1 \times (-8)$

$k = 2$: 7×2

$k = 1$: $7 \times (-1)$

$k = 0$: 1×1

Figure 3: The lattice of subspaces of $\mathbf{F}_2^3$: the zero subspace, seven lines, seven planes and the whole space; each plane contains three lines. On the right, for each dimension k, the number $\begin{bmatrix} 3 \\ k \end{bmatrix}_2$ of subspaces and the value $(-1)^k q^{\binom{k}{2}}$ of the Möbius function between 0 and a subspace of dimension k, for $q = 2$. The sum $1 - 7 + 14 - 8$ is zero: this is the identity of page 22 for $q = 2$.

Theorem 6.1 (Quillen, notebook 1970-15, pp. 21–22)**.** *Let R be a commutative ring and $q \in R$. Define $[\begin{smallmatrix} n \\ k \end{smallmatrix}]_q$ by $[\begin{smallmatrix} n \\ 0 \end{smallmatrix}] = 1$, $[\begin{smallmatrix} 0 \\ k+1 \end{smallmatrix}] = 0$ and $[\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix}] = [\begin{smallmatrix} n \\ k \end{smallmatrix}] + q^{k+1} [\begin{smallmatrix} n \\ k+1 \end{smallmatrix}]$. Then, for all $t \in R$,*

$$\prod_{i=0}^{n-1} (1 + q^i t) = \sum_{k=0}^{n} q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q t^k, \qquad \text{and} \qquad \sum_{k=0}^{n} (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q = 0 \quad (n \geq 1).$$

For $n = 3$, this is the identity of page 22.

Proof. By induction on n, for all t at once. Write $\prod_{i \leq n}(1 + q^i t) = (1 + t) \prod_{i<n}(1 + q^i (qt))$ and apply the induction hypothesis to qt. The coefficient of t^{k+1} in the product with $1 + t$ is $q^{\binom{k}{2}+k} [\begin{smallmatrix} n \\ k \end{smallmatrix}] + q^{\binom{k+1}{2}+k+1} [\begin{smallmatrix} n \\ k+1 \end{smallmatrix}]$, that is $q^{\binom{k+1}{2}} [\begin{smallmatrix} n+1 \\ k+1 \end{smallmatrix}]$ by Pascal's recurrence, since $\binom{k+1}{2} = \binom{k}{2} + k$ and $[\begin{smallmatrix} n \\ n+1 \end{smallmatrix}] = 0$. For $t = -1$ and $n \geq 1$, the factor $1 + q^0 \cdot (-1)$ vanishes. □

Verification in Lean. The coefficients are `qBinom`, with `qBinom_eq_zero_of_lt`; the binomial theorem is `prod_one_add_eq_sum`, the alternating sum is `sum_alternating_eq_zero`, and the check of the page is `quillen_check_rank_three`, in `Notebook1970_15.lean`. mathlib has no Gaussian binomial coefficients; the file defines them.

Not formalized: the counting of subspaces of $\mathbf{F}_q^n$ by the $[\begin{smallmatrix} n \\ k \end{smallmatrix}]_q$, and Möbius inversion over isotropy types. The proposition of page 14 (a pro-ℓ-group has cohomology of finite type if and only if it has an open subgroup of finite cohomology) would require the cohomology of profinite groups, absent from mathlib; it is not attempted.

Literature. found in the literature: P. Hall (1936); Rota, "On the foundations of combinatorial theory I" (1964); Stanley, *Enumerative Combinatorics* I, 3.10. The statement is classical; the edition does not present it as new.

7 Notebook 1971-4: the group completion of a commutative monoid is a flat functor[7]

Context. notebook 1971-4 (20 March – 11 April 1971) studies torsors under a small category $\mathcal{C}$ over a space or a topos, with a view to Mather's theorem. On PDF page 24 (its "8", in the second pagination), Quillen says that a functor $u : \mathcal{C}_1 \to \mathcal{C}_2$ is *strongly cofinal* if $(u_!, u^*)$ is a morphism of presheaf topoi, which amounts to requiring that, for every object Y of $\mathcal{C}_2$, the functor $X \mapsto \mathrm{Hom}(Y, uX)$ be a pro-object of $\mathcal{C}_1$, in other words that its category of elements be cofiltered. He notes that such a u induces isomorphisms on the $R^q \varprojlim$. On page 25 (its "9"), he tests the definition on the universal map from a commutative monoid M to its group completion G, viewed as a functor between one-object categories: the condition becomes that the category whose objects are the elements of G, and whose arrows are given by translations by M, is cofiltered, and he checks the two axioms in a few lines.

Intuition. An arrow $g \to g'$ is an $m \in M$ such that $mg = g'$. Two elements of G can be written with a common denominator, $g_i = m_i\, b^{-1}$, and are thus translates of b^{-1}: this is the axiom on products. Two parallel arrows $m_1, m_2 : g \to g'$ have the same image in G; by the construction of the Grothendieck group, $c\, m_1 = c\, m_2$ for some $c \in M$, and precomposing with the arrow $c : c^{-1} g \to g$ equalizes them: this is the axiom on equalizers.

Theorem 7.1 (Quillen, notebook 1971-4, p. 25)**.** *Let M be a commutative monoid, $G(M)$ its Grothendieck group and $u : M \to G(M)$ the canonical map. The induced functor between one-object categories is representably flat: for the unique object X, the category X/u is cofiltered. In particular, it is final.*

Proof. The objects of X/u are the elements of $G(M)$, and an arrow $g \to g'$ is an $m \in M$ such that $u(m)\, g = g'$. The category is nonempty (it contains 1).

Every $x \in G(M)$ satisfies $u(b)\, x = u(a)$ for some $a, b \in M$. For g_1, g_2, choose $u(b_i)\, g_i = u(a_i)$ and put $g = u(b_1 b_2)^{-1}$; then $u(b_2 a_1)\, g = g_1$ and $u(b_1 a_2)\, g = g_2$.

If $m_1, m_2 : g \to g'$, then $u(m_1)\, g = u(m_2)\, g$, so $u(m_1) = u(m_2)$, so $c\, m_1 = c\, m_2$ for some $c \in M$, since $G(M)$ is the localization of M at all of M. The arrow $c : u(c)^{-1} g \to g$ equalizes m_1 and m_2.

A representably flat functor is final (mathlib, `final_of_representablyFlat`). □

Verification in Lean. In `Notebook1971_4.lean`, the functor is `toGroupFunctor`, induced by `Algebra.GrothendieckGroup.of`; the theorem is `representablyFlat_toGroupFunctor`, in the sense of mathlib's `CategoryTheory.RepresentablyFlat` (all the categories `StructuredArrow X F` cofiltered), and `final_toGroupFunctor`. Quillen's two axioms are isolated as statements about the Grothendieck group: `exists_common` (products) and `exists_mul_eq_mul` (equalizers), with `exists_mul_eq_of`. The monoid is written multiplicatively, as on the page.

Not formalized: the equivalence of the definitions on page 24 (morphism of topoi, pro-objects, cofiltered categories of elements), the remark on the $R^q \varprojlim$, and the statements on torsors, which mathlib does not have over a small category.

Literature. folklore of group completion, found in the literature: flat functors and filtered categories of elements, SGA 4, exposés I and IV; Mac Lane–Moerdijk, *Sheaves in Geometry and Logic* (1992), VII.6. mathlib has `RepresentablyFlat` and the Grothendieck group, but not this example. It is proposed for mathlib, and the edition does not present it as new.

[7] Proof checked: `lean/Quillen/Notebook1971_4.lean`.

8 Notebook 1971-11: automorphisms of order p of an abelian p-group that are trivial on the elements of order p ♦[8]

Context. In 1971, Quillen studies the cohomology of finite groups and the role of their elementary abelian subgroups. notebook 1971-11 contains, from page 68 on, a series dated 24 August 1971, titled "p odd", annotated in the margin as a revision of Thompson's theorems. One reads there a lemma, then two propositions on the maximal normal elementary abelian subgroups of a p-group, which use it (pp. 69–71).

Notation. For an abelian group B, written additively, and a prime p, we write $\Omega_1(B)$ for the subgroup of elements b such that $pb = 0$. An *abelian p-group* is an abelian group in which every element is killed by a power of p.

The page. On page 68, Quillen states the following lemma: if θ is an automorphism of an abelian p-group B, with $\theta^p = 1$, which is the identity on $\Omega_1(B)$, then $\theta - 1$ sends B into $\Omega_1(B)$. His proof, on page 69, passes to the dual group $\widehat{B}$, writes $\theta = 1 + N$ with N having image in $p\widehat{B}$, and compares the terms of the binomial expansion of $(1 + N)^p = 1$; a first attempt, by a p-adic filtration, is crossed out. The hypothesis $p \geq 3$ serves to make the last term N^p small enough. Page 69 then applies the lemma to an element of order p that normalizes an abelian subgroup.

Intuition. Put $\delta = \theta - 1$. The hypothesis says that δ kills $\Omega_1(B)$, and the conclusion that $p\delta = 0$. Since δ commutes with multiplication by p, it lowers the order: if $p^{k+1}b = 0$, then $p^k\delta(b) = \delta(p^k b) = 0$, since $p^k b \in \Omega_1(B)$. The powers of δ thus get smaller and smaller, and in the expansion of $(1 + \delta)^p = 1$ the term $p\delta$ can be compensated only by vanishing terms.

Theorem 8.1 (Quillen, notebook 1971-11, p. 68)**.** *Let $p \geq 3$ be a prime, B an abelian group in which every element is killed by a power of p, and θ an endomorphism of B such that $\theta^p = 1$ and $\theta(b) = b$ for all $b \in \Omega_1(B)$. Then $p(\theta(b) - b) = 0$ for all $b \in B$, that is, $(\theta - 1)B \subset \Omega_1(B)$.*

The statement is that of the page, in a slightly more general form: B is not assumed finite, and θ is an endomorphism, automatically bijective since $\theta^p = 1$.

Proof. Let $\delta = \theta - 1$. We show, by induction on k, that $p\delta(c) = 0$ for every c such that $p^k c = 0$. For $k = 0$, $c = 0$. Suppose the result true for k, and let c be such that $p^{k+1}c = 0$. By the lowering of the order, $p^k\delta(c) = 0$; the induction hypothesis applied to $\delta(c)$ gives $p\delta^2(c) = 0$. Then $\delta^2(c) \in \Omega_1(B)$, so $\delta^3(c) = 0$, then $\delta^m(c) = 0$ for all $m \geq 3$, and $p\delta^m(c) = 0$ for all $m \geq 2$.

Since $\theta = 1 + \delta$ and 1 commutes with δ, the binomial formula gives

$$c = \theta^p(c) = \sum_{m=0}^{p} \binom{p}{m} \delta^m(c) = c + p\,\delta(c) + \sum_{m=2}^{p-1} \binom{p}{m} \delta^m(c) + \delta^p(c).$$

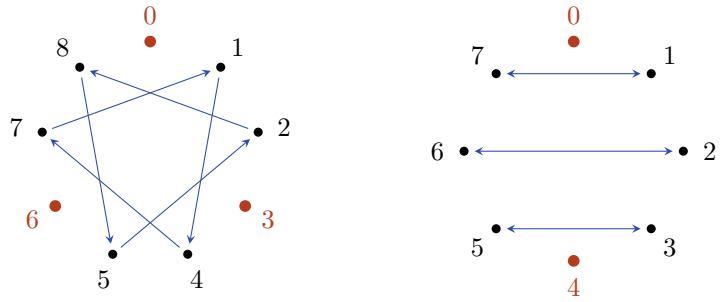
For $2 \leq m \leq p - 1$, the coefficient $\binom{p}{m}$ is divisible by p, and $p\delta^m(c) = 0$: these terms vanish. The last one, $\delta^p(c)$, vanishes because $p \geq 3$. What remains is $p\delta(c) = 0$. □

The proof is more direct than that of the page: it does not pass to the dual and proceeds by induction on the order of the elements. The hypothesis $p \geq 3$ serves at the same place: for $p = 2$, the last term is $\delta^2(c)$, which the induction cannot cancel.

[8] Proof checked: `lean/Quillen/Notebook1971_11.lean`.

Counterexample 8.2 (the edition's statement)**.** *For $p = 2$, the lemma is false. On $B = \mathbf{Z}/8$, let $\theta = -1$. Then $\theta^2 = 1$, and θ fixes $\Omega_1(B) = \{0, 4\}$, since $-4 = 4$. But $2(\theta(1) - 1) = -4 \neq 0$.*

On $\mathbf{Z}/4$, on the other hand, $\theta = -1$ satisfies the conclusion ($2 \cdot (-2) = 0$): the first cyclic counterexample requires order 8. An exhaustive search, over the groups $\mathbf{Z}/9$, $\mathbf{Z}/27$, $\mathbf{Z}/9 \times \mathbf{Z}/3$, $\mathbf{Z}/9 \times \mathbf{Z}/9$, $\mathbf{Z}/27 \times \mathbf{Z}/3$ for $p = 3$ and $\mathbf{Z}/25$ for $p = 5$, found no automorphism contradicting the lemma; for $p = 2$, it finds some on $\mathbf{Z}/8$ and $\mathbf{Z}/8 \times \mathbf{Z}/2$, and none on $\mathbf{Z}/4$, $\mathbf{Z}/4 \times \mathbf{Z}/2$, $\mathbf{Z}/4 \times \mathbf{Z}/4$. The title "$p$ odd" of the page is therefore necessary (figure 4). The hypothesis that B is a p-group is necessary too: the ring $\mathbf{Z}[\zeta_p]$ of cyclotomic integers, with θ multiplication by ζ_p, has $\Omega_1 = 0$ and $\theta^p = 1$, without $p(\theta - 1)$ being zero.



$p = 3$, $B = \mathbf{Z}/9$, $\theta = 4$ $\qquad$ $p = 2$, $B = \mathbf{Z}/8$, $\theta = -1$

Figure 4: The action of θ (arrows) and the subgroup $\Omega_1(B)$ (in red). On the left, an example of the edition: multiplication by 4 on $\mathbf{Z}/9$ satisfies $\theta^3 = 1$, fixes $\Omega_1(B) = \{0, 3, 6\}$ and cyclically permutes $1, 4, 7$ and $2, 8, 5$; $\theta - 1$ is multiplication by 3, with values in $\Omega_1(B)$, as the theorem requires. On the right, counterexample 8.2: $\theta = -1$ fixes $\Omega_1(B) = \{0, 4\}$, but $\theta(1) - 1 = 6$ is not in $\Omega_1(B)$.

Verification in Lean. The file `Notebook1971_11.lean` proves Theorem 8.1 under the name `nsmul_sub_one_eq_zero`, for θ in the ring `AddMonoid.End B` of endomorphisms of B, and under the name `addEquiv_nsmul_sub_one_eq_zero` for an automorphism `B ≃+ B` whose p-th iterate is the identity, as on the page. The lowering of the order is `sub_one_nsmul_eq_zero_of_nsmul`, the binomial expansion evaluated at a point is `pow_apply_eq_sum`. Counterexample 8.2 is `zmod8_neg`, checked by computation. The axioms used are the three usual axioms of mathlib.

Literature. the search did not locate a printed reference stating this lemma in this form. It is of the type of lemmas that serve Thompson's replacement theorems, which the page says it revises, and the study of elementary abelian subgroups in Quillen's 1971–1978 work on the cohomology of finite groups; the expositions of these theorems in textbooks on finite groups have not been combed. The statement is elementary, and the edition does not present it as new.

The quotient of a p-central p-group by Ω_1[9]

Notation. For a group P, we write $\Omega_1(P)$ for the subgroup generated by the elements x such that $x^p = 1$; it is always normal. We say that P is *p-central* if every element of order p of P is central, that is, if $\Omega_1(P) \subset Z(P)$. In this case, $\Omega_1(P)$ is abelian of exponent p and consists exactly of the elements x such that $x^p = 1$.

[9] Proof checked: `lean/Quillen/Notebook1971_11Central.lean`.

The page. The series of 24 August 1971 ends, on pages 70–71 of notebook 1971-11, with a proposition: for p odd, if P is a p-central p-group, the quotient $P/\Omega_1(P)$ is also p-central. Quillen's proof puts $A = \Omega_1(P)$ and chooses a subgroup $B \supset A$, normal in P, such that B/A is a maximal normal elementary abelian subgroup of P/A. Raising to the p-th power sends B into A; it induces an injective map from B/A to A, compatible with conjugation since A is central, from which he deduces that B/A is central in P/A. He concludes with the preceding proposition (pp. 69–70): a maximal normal elementary abelian subgroup is a maximal elementary abelian subgroup (p odd), so that B/A contains all the elements of order p of P/A. It is this intermediate proposition that uses the lemma of Theorem 8.1.

Intuition. An element x whose class modulo A has order dividing p satisfies $x^p \in A$. For $g \in P$, the conjugate $u = gxg^{-1}$ has the same p-th power as x, since x^p is central. If u and x commute modulo A, the group they generate has class at most 2, and for p odd raising to the p-th power is a homomorphism there: $(ux^{-1})^p = u^p x^{-p} = 1$, so the commutator $gxg^{-1}x^{-1}$ is in A. It remains to see that u and x commute modulo A; this is the statement itself in a maximal subgroup containing x, whence an induction on the order.

Theorem 8.3 (Quillen, notebook 1971-11, pp. 70–71)**.** *Let p be an odd prime and P a finite p-group in which every element z with $z^p = 1$ is central. Then every element y of $P/\Omega_1(P)$ with $y^p = 1$ is central; in other words, $\Omega_1(P/\Omega_1(P)) \subset Z(P/\Omega_1(P))$.*

Proof. Put $A = \Omega_1(P)$. Since $A = \{x \in P \mid x^p = 1\}$, we must show: if $(x^p)^p = 1$, then $(gxg^{-1}x^{-1})^p = 1$ for all $g \in P$. We argue by induction on the order of P; the hypothesis passes to subgroups, since an element central in P is central in every subgroup containing it.

If x generates P, the group is commutative and the commutator is trivial. Otherwise, $\langle x \rangle$ is contained in a maximal subgroup M, normal because P is nilpotent. The conjugate $u = gxg^{-1}$ is in M and satisfies $u^p = gx^pg^{-1} = x^p$, whence $(u^p)^p = 1$. The induction hypothesis, applied in M to the elements u and x^{-1}, gives $d^p = 1$ for $d = x^{-1}uxu^{-1}$; thus d is central. We have $x^{-1}u = d\,u\,x^{-1}$, and an immediate induction gives, for all n,

$$(ux^{-1})^n = u^n\, x^{-n}\, d^{\binom{n}{2}}.$$

For $n = p$ odd, p divides $\binom{p}{2}$, so $d^{\binom{p}{2}} = 1$ and $(ux^{-1})^p = u^p x^{-p} = 1$. Now $ux^{-1} = gxg^{-1}x^{-1}$.

Finally, if $y = xA$ satisfies $y^p = 1$, then $x^p \in A$, that is $(x^p)^p = 1$, and for all g the commutator $gxg^{-1}x^{-1}$ is in A: y commutes with gA. □

The proof keeps Quillen's first step (raising to the p-th power is an injective homomorphism compatible with conjugation, which forces the commutators into A), but replaces the appeal to the proposition of pages 69–70 by an induction on the order of P. It therefore does not use the lemma of Theorem 8.1. The hypothesis p odd serves at the same place as in Quillen: the divisibility of $\binom{p}{2}$ by p, which makes raising to the p-th power a homomorphism in class 2.

Verification in Lean. The file `Notebook1971_11Central.lean` defines $\Omega_1(P)$ under the name `omega1`, as the subgroup generated by the elements x such that $x^p = 1$, and shows that it is normal (`omega1_normal`) and, under the hypothesis of p-centrality, that it consists exactly of these elements (`mem_omega1_iff`). The formula $(uv)^n = u^n v^n d^{\binom{n}{2}}$ is `mul_pow_eq_mul_pow_mul_pow_choose`, and its case $n = p$ odd is `mul_pow_eq_of_commutator_pow`. The induction on the order is `conj_commutator_pow_eq_one_aux`, its statement `conj_commutator_pow_eq_one`. Theorem 8.3 is `pCentral_quotient`, for elements, and `omega1_quotient_le_center`, in the form

of the page $\Omega_1(P) \leq Z(P) \Rightarrow \Omega_1(P/\Omega_1(P)) \leq Z(P/\Omega_1(P))$. The axioms used are the three usual axioms of mathlib.

Not proved here is the proposition of pages 69–70 (a maximal normal elementary abelian subgroup of a p-group, p odd, is a maximal elementary abelian subgroup), through which Quillen's proof passes and which uses the lemma of Theorem 8.1.

Literature. the statement is known before the notebook: it is the theorem that J. González-Sánchez and Th. Weigel (2011, introduction) call Blackburn–Buckley–Thompson. It is published in this form by J. Buckley, "Finite groups whose minimal subgroups are normal", *Math. Z.* 116 (1970), 15–17, and follows from a result of Thompson that B. Huppert reproduces as early as 1967, with a proof attributed to Blackburn (*Endliche Gruppen* I, Springer, III, Hilfssatz 12.2). Theorem B of González-Sánchez and Weigel generalizes it to p^i-central pro-p-groups of height k. The notebook reproves it in 1971, as a lemma in a revision of Thompson's theorems; only the proof by induction given here is the edition's.

9 Notebook 1983-4: the cohomological equation on the torus and Diophantine numbers[10]

Context. In spring 1983, notebook 1983-4 studies the Connes algebra of the Kronecker foliation of irrational slope θ on the torus T^2 and its projective modules. On pages 71–72 (its pages 800–801), Quillen computes the Picard group of the topos of leaves of this foliation: the line bundles there are line bundles on T^2 equipped with a connection that is flat along the leaves, and the group sought is $H^1(T^2, \mathcal{O}^*)$, where $\mathcal{O}$ is the sheaf of smooth functions constant along the leaves.

The page. The first step is $H^1(T^2, \mathcal{O})$, computed by the fine resolution given by the operator $D = \theta\partial_x - \partial_y$ on $C^\infty(T^2)$. On the exponential $e^{2\pi i(mx+ny)}$, D acts by the scalar $2\pi i(\theta m - n)$, which vanishes only at $(m, n) = (0, 0)$ since θ is irrational. Quillen deduces that D is invertible off the constants, hence that H^0 and H^1 are both equal to $\mathbf{C}$, then, by the exponential sequence, that the Picard group is $\mathbf{C}^*/\{e^{2\pi in\theta}\} \times \mathbf{Z}$.

Intuition. Inverting D on the exponentials is not enough: the series

$$\sum b_{m,n}\,(2\pi i(\theta m - n))^{-1} e^{2\pi i(mx+ny)}$$

must still define a smooth function. The coefficients of a smooth function decay faster than any power of $1 + |m| + |n|$; if the small divisors $|\theta m - n|$ are bounded below only by a power of $|m|^{-1}$, the division costs a fixed power and rapid decay survives. If on the contrary θ is very well approximated by rationals, one can place tiny coefficients $b_{m,n}$ exactly at the points where $\theta m - n$ is smaller still, and the quotient no longer decays. This is the phenomenon of small divisors, and it is the hypothesis that the page does not state.

Statement. *The sequences considered are indexed by $\mathbf{Z}^2$ and complex-valued. A sequence a is rapidly decreasing if, for every integer k, the quantity $|a_{m,n}|\,(1 + |m| + |n|)^k$ is bounded. A real number θ is Diophantine if there exist $C > 0$ and an integer N such that $|\theta m - n| \geq C\,|m|^{-N}$ for all integers $m \neq 0$ and n. The factor 2π is omitted: D acts by $a_{m,n} \mapsto i(\theta m - n)\,a_{m,n}$.*

[10] Proof checked: `lean/Quillen/Notebook1983_4.lean`.

Theorem 9.1 (the edition's statement; missing hypothesis of notebook 1983-4, pp. 71–72)**.** *Let θ be an irrational real number. The operator $a \mapsto (i(\theta m - n)\, a_{m,n})_{m,n}$ sends rapidly decreasing sequences to rapidly decreasing sequences whose $(0,0)$ term is zero, and it is surjective onto the latter if and only if θ is Diophantine. More precisely: (1) if θ is Diophantine (without any irrationality hypothesis) and b is rapidly decreasing with $b_{0,0} = 0$, the sequence $a_{m,n} = b_{m,n}/(i(\theta m - n))$, $a_{0,0} = 0$, is rapidly decreasing and satisfies $i(\theta m - n)\, a_{m,n} = b_{m,n}$; (2) if θ is irrational and not Diophantine, there exists a rapidly decreasing sequence b, with $b_{0,0} = 0$, that has no rapidly decreasing preimage. In particular, the conclusion of the page holds for θ Diophantine, and fails for a Liouville number.*

Proof. Write $w_{m,n} = 1 + |m| + |n|$ and $\sigma_{m,n} = \theta m - n$. We have $|\sigma_{m,n}| \leq (|\theta| + 1)\, w_{m,n}$, whence the first assertion.

(1) There exist $c > 0$ and N such that $|\sigma_{m,n}|\, w_{m,n}^N \geq c$ for $(m,n) \neq (0,0)$: if $m \neq 0$, this is the hypothesis, since $|m| \leq w_{m,n}$; if $m = 0$, we have $|\sigma_{0,n}| = |n| \geq 1$. Then $|a_{m,n}|\, w^k \leq |b_{m,n}|\, w^{k+N}/c$, which is bounded. The lower bound involves only $|m|$, and $|m| \leq w$: the dependence on n is no obstacle in this direction.

(2) The negation of the hypothesis provides, for each j, a pair $q_j = (m_j, n_j)$ with $m_j \neq 0$ and $|\sigma_{q_j}| < (j+1)^{-1} |m_j|^{-2j}$; irrationality gives $\sigma_{q_j} \neq 0$. Put $b_q = \sigma_q / \sqrt{|\sigma_q|}$ if q is one of the q_j, and $b_q = 0$ otherwise; thus $b_{0,0} = 0$ and $|b_{q_j}| = \sqrt{|\sigma_{q_j}|} < |m_j|^{-j}$. Here n comes in: since $|\sigma_{q_j}| < 1$, we have $|n_j| \leq |\theta|\, |m_j| + 1$, whence $w_{q_j} \leq (3 + |\theta|)\, |m_j|$. For $j \geq k$, we get $|b_{q_j}|\, w_{q_j}^k \leq (3 + |\theta|)^k |m_j|^{k-j} \leq (3 + |\theta|)^k$; the indices $j < k$ are finitely many. Hence b is rapidly decreasing. If a were a preimage of it, we would have $|a_{q_j}| = |\sigma_{q_j}|^{-1/2} > \sqrt{j+1}$, and a would not even be bounded. □

Proposition 9.2. *A Liouville number is not Diophantine. For such a θ, there is therefore a rapidly decreasing sequence with zero $(0,0)$ term that is not in the image of D.*

Proof. Let $C > 0$ and N. For infinitely many denominators b, there exists a with $|\theta - a/b| < b^{-(N+2)}$; taking $b \geq 1/C$, we obtain $|\theta b - a| < b^{-(N+1)} \leq C\, b^{-N}$, which contradicts the lower bound. A Liouville number is irrational, and Theorem 9.1 applies. □

Remark. For θ non-Diophantine, the image of D in the smooth functions of mean zero is dense but not closed: the quotient $H^1(T^2, \mathcal{O})$ computed by the resolution is infinite-dimensional and non-Hausdorff. The computation of the Picard group that follows in the notebook therefore rests on the Diophantine hypothesis. The set of Diophantine reals has full measure, so that the conclusion of the page holds for almost every θ.

Verification in Lean. The file `Notebook1983_4.lean` works at the level of sequences on $\mathbf{Z} \times \mathbf{Z}$, without Fourier analysis. It defines the weight `weight`, rapid decay `IsRapidDecay`, the predicate `IsDiophantine` and the symbol `symbol`. The forward direction is `IsDiophantine.isRapidDecay_div` and `IsDiophantine.symbol_mul_div`, from the lower bound `IsDiophantine.lower_bound`; the stability of rapid decay is `IsRapidDecay.symbol_mul`. The converse is `exists_not_mem_range_of_not_isDiophantine`, the equivalence is `surjective_iff_isDiophantine`, and the case of Liouville numbers is `not_isDiophantine_of_liouville` and `exists_not_mem_range_of_liouville`, from mathlib's `Liouville` predicate. mathlib has neither rapidly decreasing sequences on $\mathbf{Z}^2$ nor Diophantine numbers in this sense; the file defines them.

Not formalized: the identification of smooth functions on T^2 with rapidly decreasing sequences of their Fourier coefficients, the resolution of the sheaf $\mathcal{O}$, the non-Hausdorff character of H^1 in the Liouville case, the full measure of the Diophantine numbers, and the computation of the Picard group.

Literature. found in the literature: the phenomenon of small divisors and the cohomological equation on the torus are classical (V. I. Arnold, *Geometrical Methods in the Theory of Ordinary Differential Equations*, supplementary chapters; M. Herman, Publ. Math. IHÉS 49 (1979); A. Katok, *Combinatorial constructions in ergodic theory and dynamics*, AMS (2003)). For the foliated cohomology of the Kronecker foliation, which is $\mathbf{C}$ in degree 1 if and only if θ is Diophantine and infinite-dimensional non-Hausdorff otherwise: A. El Kacimi Alaoui, Compositio Math. 49 (1983); A. Haefliger, J. Differential Geom. 15 (1980). The statement is classical; the edition does not present it as new and only points out the hypothesis that the page does not state.

10 Notebook 1983-10: the trace of the odd powers of a difference of two idempotents ♦[11]

Context. notebook 1983-10 works on index formulas in cyclic cohomology, the theory that Connes was developing at the time and to which Quillen would devote several papers. The index of a Toeplitz operator, or of a graded operator, is written there as the trace of a sufficiently large odd power of an operator built from an idempotent.

The page. On page 31 (its page 154), Quillen observes that his two index formulas, the Toeplitz form and the graded form, reduce to the same type of expression: the trace of an odd power of a difference $e_1 - e_2$ of two idempotents. He proposes to study $\operatorname{tr}((e_1 - e_2)^{2n+1})$ and to try to show that it is an integer independent of n. The page poses the question; it does not answer it.

Intuition. In infinite dimension, $e_1 - e_2$ is not trace class in general, and only a sufficiently large power is; the question is then deep. In finite dimension, all the traces exist, and the expected answer is $\operatorname{tr} e_1 - \operatorname{tr} e_2$, the difference of the ranks when working over a field of characteristic zero. The classical computation puts $A = P - Q$ and $B = P + Q - 1$: these two matrices anticommute and $A^2 + B^2 = 1$ (figure 5). For m odd, $\operatorname{tr}(A^m B^2)$ equals its opposite, by cyclicity of the trace and anticommutation, whence $\operatorname{tr}(A^m) = \operatorname{tr}(A^{m+2})$ *provided one can divide by* 2. One might think that characteristic 2 is an exception. It is not.

Theorem 10.1 (the edition's statement; answer, in finite dimension, to the question of notebook 1983-10, p. 31)**.** *Let R be a commutative ring and P, Q two idempotent square matrices with entries in R. For every integer $k \geq 0$,*

$$\operatorname{tr}\big((P - Q)^{2k+1}\big) = \operatorname{tr} P - \operatorname{tr} Q.$$

Proof. Put $C = (P - Q)^2 = P + Q - PQ - QP$. Expanding, with $P^2 = P$ and $Q^2 = Q$,

$$PC = P - PQP = CP, \qquad QC = Q - QPQ = CQ :$$

[11] Proof checked: `lean/Quillen/Notebook1983_10.lean`.

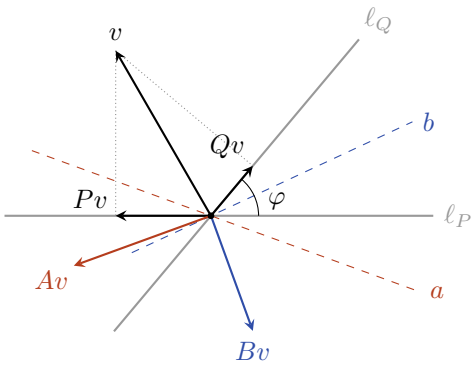


Figure 5: Two orthogonal projectors P and Q of the Euclidean plane, onto lines ℓ_P and ℓ_Q making an angle φ (here $50°$). The matrix $A = P - Q$ is $\sin\varphi$ times the orthogonal reflection in the axis a, and $B = P+Q-1$ is $\cos\varphi$ times the reflection in the axis b, the bisector of ℓ_P and ℓ_Q. The two axes make an angle of $45°$: the product of the two reflections is a quarter turn, whence $AB = -BA$, and $A^2 + B^2 = (\sin^2\varphi + \cos^2\varphi)\cdot 1 = 1$. Here $\operatorname{tr} A^{2k+1} = 0 = \operatorname{tr} P - \operatorname{tr} Q$.

C commutes with P and Q, hence so do its powers. It follows that

$$(P-Q)\,C = (P-Q) - (PQP - QPQ).$$

Let D be a matrix commuting with P and Q. By cyclicity of the trace, $\operatorname{tr}(DPQP) = \operatorname{tr}(PDPQ) = \operatorname{tr}(DP^2Q) = \operatorname{tr}(DPQ)$; likewise $\operatorname{tr}(DQPQ) = \operatorname{tr}(DQP)$, and $\operatorname{tr}(DQP) = \operatorname{tr}(PDQ) = \operatorname{tr}(DPQ)$. Hence

$$\operatorname{tr}(DPQP) = \operatorname{tr}(DQPQ).$$

Finally $(P-Q)^{2k+1} = C^k(P-Q)$, and

$$(P-Q)^{2k+3} = C^k(P-Q)\,C = (P-Q)^{2k+1} - \big(C^kPQP - C^kQPQ\big).$$

The trace of the term in parentheses is zero, by the above with $D = C^k$. The traces of the odd powers of $P-Q$ are therefore all equal to $\operatorname{tr}(P-Q) = \operatorname{tr} P - \operatorname{tr} Q$. □

The proof uses only idempotence, the commutation of C with P and Q, and the cyclicity of the trace; it never divides by 2. The computation by anticommutation shows $2\operatorname{tr}(A^mB^2) = 0$; the proof above shows directly $\operatorname{tr}(A^mB^2) = 0$, since $A^mB^2 = C^k(PQP - QPQ)$ for $m = 2k+1$. An exhaustive computation over all pairs of idempotents of size 2 and 3 over $\mathbf{F}_2$, of size 2 over $\mathbf{Z}/4$, and over $\mathbf{F}_3$, $\mathbf{F}_5$, $\mathbf{Z}/9$, for $k = 1, 2, 3$, confirms the identity, in characteristic 2 as elsewhere.

Verification in Lean. The file `Notebook1983_10.lean` proves Theorem 10.1 under the name `trace_sub_pow_odd`, for matrices indexed by an arbitrary finite type over an arbitrary commutative ring, idempotence being mathlib's predicate `IsIdempotentElem`. The steps are `sq_sub_commute_left` and `sq_sub_commute_right` (commutation of C), `sub_mul_sq_sub` (the identity $(P-Q)C = (P-Q) - (PQP-QPQ)$) and `trace_mul_PQP`. Mathlib does not contain this identity: a pattern search on `Matrix.trace` and `IsIdempotentElem` returns nothing. The axioms used are the three usual axioms of mathlib.

Not proved here: the infinite-dimensional version, where $P-Q$ is not trace class and only a sufficiently large odd power is, which is the real question of the page; the interpretation of the common value as the index of a Fredholm operator.

Literature. the infinite-dimensional version is published: J. Avron, R. Seiler and B. Simon

[1] prove, for two orthogonal projections of a Hilbert space whose difference has a trace-class odd power $(P-Q)^{2n+1}$, that the index of the pair equals $\mathrm{tr}((P-Q)^{2n+1})$, and hence does not depend on n; their proof rests in particular on the anticommutation of $P-Q$ and $P+Q-1$. The finite-dimensional case is its simplest example. The form with no hypothesis on the characteristic has not been found in print; it is elementary, and the edition does not present it as new. It is not a lead.

Statement on the use of artificial intelligence

The statements proved here are read from transcriptions of the notebooks produced by Anthropic language models (Claude) under editorial rules written by the author; these transcriptions have not been fully reviewed by a person. The formal proofs were written in Lean 4 by Claude Opus 5.5, under the author's direction, and Lean mechanically checks their correctness against the mathlib library. The same model drew from them the written-out proofs of this document. The author reviewed the choice of statements and the text, and takes full responsibility for them. Lean guarantees that a statement holds under the hypotheses as written. It guarantees neither that the statement faithfully renders the page, nor that it is new.

References

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